CSC1042 Computer Network II

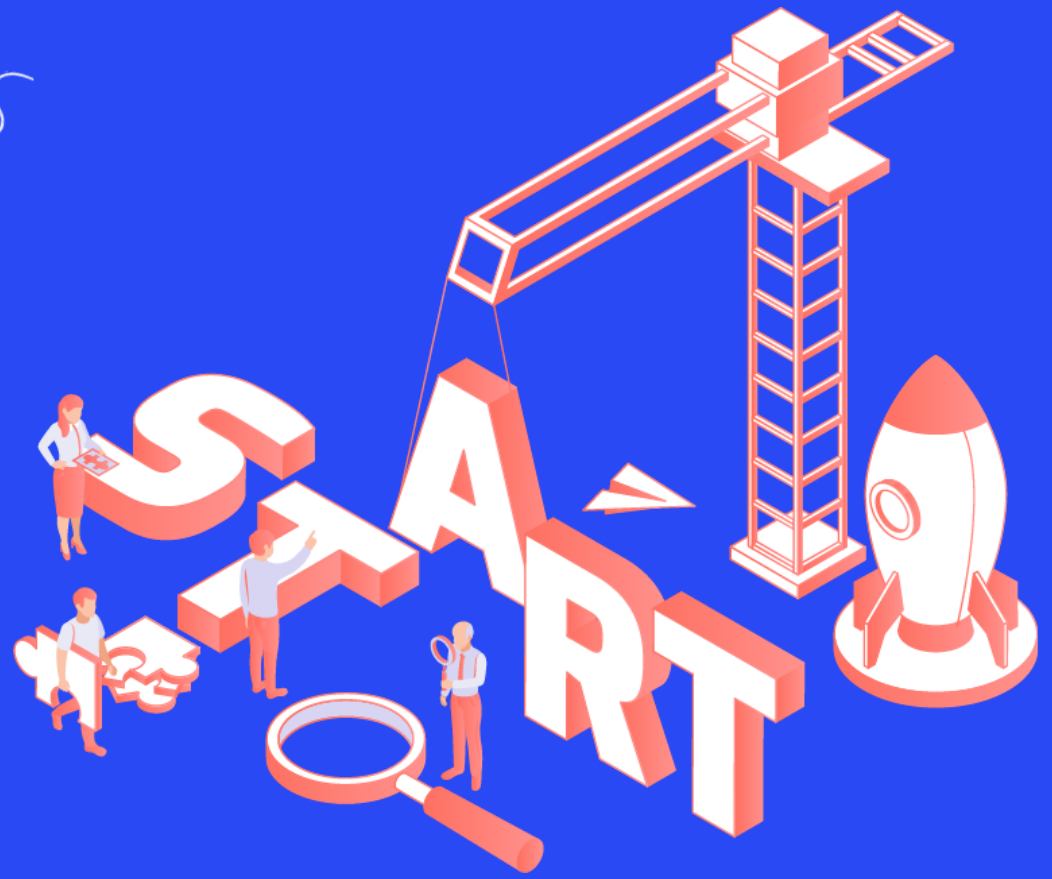
Other Routing Protocols

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Learning outcomes

- Give examples of routing protocols that use hybrid approaches
- Describe the characteristics of these protocols

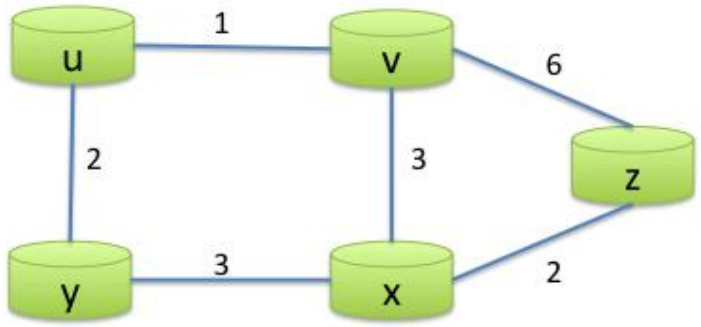


Enhanced Interior Gateway Routing Protocol (EIGRP)

- Layer 3 protocol with ID 88
- Originally, it was Cisco proprietary
  - Now it is an open standard
- A hybrid interior gateway protocol
  - It has characteristics of both distance-vector and link-state protocols
  - It sends distance-vector updates containing information about networks along with the distance to reach them
  - It builds neighbor and topology tables and send routing updates only when topology change occurs
  - It sends Hello messages to maintain router neighbour relationship
- It uses multicast to send routing information

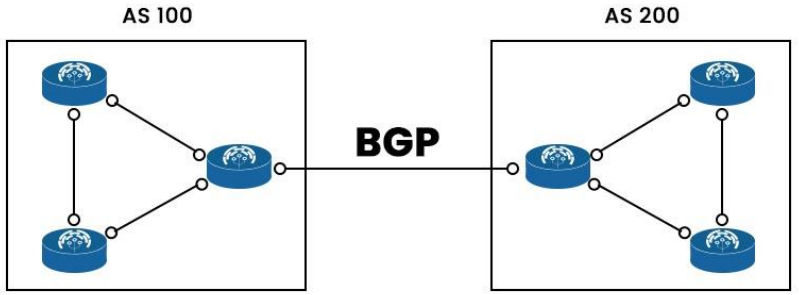
Enhanced Interior Gateway Routing Protocol (EIGRP)

- It uses 5 “K” constants to determine the metric
  - Bandwidth, delay, load, reliability, MTU
  - Default bandwidth + delay
- It selects the route with the lowest cumulative metric
- It can also load balance traffic across paths
- It converges very fast (seconds)
- It uses the concept of Autonomous System
  - Routers should know to which AS they belong to
  - Only border routers (connecting multiple AS) can belong to multiple AS



Border Gateway Protocol (BGP)

- It is a Layer 7 protocol
  - It uses unicast to send routing information through TCP port 179
- It's an exterior routing protocol
  - It routes between AS
  - It is the only one used on the Internet



- It is a variation of distance vector protocol – it's called Path Vector Protocol
  - It views an entire AS as a hop
  - BGR routers- called BGP peers- are manually configured
- BGP is always used between ISPs

Border Gateway Protocol (BGP)

- The most complex routing protocol to implement
- It uses several criteria for best path selection, including:
  - Weight of the interface
  - Source of the path
  - Path length
  - Preferred path
  - Preferred AS entry point
- If used incorrectly, a private organization could accidently become a public “transit network” between two ISPs
- Very slow convergence by default
- The focus of BGP design and implementation is on security and scalability

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THANK YOU